Jangho Lee

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PROFESSIONAL PROFILES

- Google Scholar: https://scholar.google.com/citations?user=wBEE2YAAAAAJ
- ORCID: https://orcid.org/0000-0002-8942-1092
- LinkedIn: https://www.linkedin.com/in/jholee92/
- GitHub: https://github.com/jangholee92

RESEARCH INTERESTS

Urban Natural Disaster; Nature Based Solution; Climate Informatics; Downscaling; Machine Learning; Deep Learning; Remote Sensing; Statistical Meteorology; Urban Climate; Climate Impact; Land Modelling

EDUCATION

TEXAS A&M UNIVERSITY ***2018-2023***

Doctor of Philosophy *in Atmospheric Science*
- Advisor: Dr. Andrew Dessler

SEOUL NATIONAL UNIVERSITY ***2011-2018***

Bachelor of Science *in Earth and Environmental Science*
- Advisor: Dr. Kwang-Yul Kim

RESEARCH EXPERIENCE & PROFESSIONAL APPOINTMENT

UNIVERSITY OF ILLINOIS CHICAGO ***2023-Present***

Postdoctoral Researcher
- CROCUS Artificial Intelligence and Digital Twins (AIDT) Project planning lead
- Directed collaboration with ANL and ORNL for E3SM & ELM simulations
- Managed the social engagement program in partnership with the Puerto Rican Agenda of Chicago
- Served as a leader of the postdoc association to lead the CROCUS meeting at UIC

TEXAS A&M UNIVERSUTY ***2018-2023***

Graduate Research Assistant
- Led the publication of multiple peer-reviewed paper on extreme climate and socioeconomic impact
- Led the Team in Cyber-Training program held at University of Maryland Baltimore County
- Served as international student representative and electives representative in Graduate Student Council

SEOUL NATIONAL UNIVERSITY ***2014-2018***

Undergraduate Intern
- Led the publication of multiple peer-reviewed paper on statistical climate, extreme temperature event, and dust source identification research and presented findings at various conferences

COMMUNITY ENGAGEMENT & OTHER EXPERIENCES

PUERTO RICAN AGENDA & GREATER CHATHAM OF CHICAGO ***2024-Present***

Scientific Advising Committee
- Facilitated a town hall meeting to lead the proposal of Tree Equity Grant of Chicago

REPUBLIC OF KOREA ARMY ***2012-2014***

Drill Sergent at Korean Army Training Center

PEER-REVIEWED PUBLICATIONS

** Corresponding Author*

2025

1. **Lee. J***., Park, S.Y., Wadhwa, A., Packman, A., Nesbitt, S. W., Sharma, A., Berkelhammer, M., Garcia, M. H., Kotamarthi, R., Hence, D., Miller, W. M (2025) Comparing Multi-Source Urban Flood Indicators: Satellite, Simulation, and Citizen-Reported Data. *Environmental Research: Water* [Accepted].
2. Kotamarthi, R*., Wang, J., Stock, J., Fytanidis, D. K., Munson, T., Niyogi, D., Muradyan, P., Jackson, R., Collis, S., Gonzalez-Meler, M. A., **Lee, J**., Sharma, A., Wang, S., Kaludi, B., Kaplan, M., Nesbitt, S. W., Martilli, A., Jacob, R., Negri, C (2025) Artificial Intelligence-Enabled Digital Twin for U.S. Cities. *Bulletin of the American Meteorological Society* [Accepted].
3. **Lee. J***. & Berkelhammer, M (2025) Evaluating the Influence of Traffic Congestion on Surface Urban Heat Island Intensity. *Geophysical Research Letters*, 52(16), e2025GL116651.
4. **Lee, J***., Berkelhammer, M., Park, S. Y., Wilson, M (2025) Analysis of Urban Flooding in Chicago Based on Crowdsourced Data: Drivers and the Need for Community-Based Mitigation Strategies. *Environmental Research: Infrastructure and Sustainability*, 5 (2), 025008
5. Cho, A*., Love, N., Cintron, R., Nicholson, J., Xu, L., Nunez-Mir, G., **Lee, J**., Berkelhammer, M., Gonzalez-Meler, M (2025) Plant Species Selection and Participatory Community Co-design are Essential in Balancing Ecosystem Services and Disservices in Urban Areas. *Environmental Research Letters*, 20, 051003.
6. **Lee, J*.** & Park, S.Y. (2025) WGAN-GP-Based Conditional GAN (cGAN) with Extreme Critic for Precipitation Downscaling in a Key Agricultural Region of the Northeast U.S. *IEEE Access–Geoscience and Remote Sensing Society Section*, 13, 46030-46041.
7. **Lee, J*.** (2025) Inferring Urban Air Temperatures from Land Surface Temperatures with the E3SM Land Model (uELM), Satellite Observations, and Measurement Campaign. *IEEE Access–Geoscience and Remote Sensing Society Section*, 13, 32564-32573.
8. **Lee, J*.** (2025) Estimating Near-Surface Air Temperature from Satellite-Derived Land Surface Temperature Using Temporal Deep Learning: A Comparative Analysis. *IEEE Access–Geoscience and Remote Sensing Society Section*, 13, 28935-28945.

2024

9. **Lee, J*.**, & Berkelhammer, M. (2024) Observational Constraints on the Spatial Effect of Greenness and Canopy Cover on Urban Heat in Major Midlatitude City. *Geophysical Research Letters*, 51(1), e2024GL110847.
10. Cho, A*., Dziedzic, N., Davis, A., Hanson, C., **Lee, J.**, Nunez-Mir, G., Gonzalez-Meler, M. A. (2024). Leaf Functional Traits Highlight Phenotypic Variation of Two Tree Species in the Urban Environment. *Frontiers in Plant Science Functional Plant Ecology*, 15, 1450723.
11. **Lee, J*.** (2024). Assessment of U.S. Urban Surface Temperature using GOES-16 and 17 Data: Urban Heat Island and Temperature Inequality. *Weather, Climate, and Society,* 16(2), 315-329.
12. **Lee, J***., Berkelhammer, M., Wilson, M. D., Love, N., & Cintron, R. (2024). Urban Land Surface Temperature Downscaling in Chicago: Addressing Ethnic Inequality and Gentrification. *Remote Sensing*, 16(9), 1639.
13. **Lee. J***., & Hu, M. (2024). Effect of Environmental and Socioeconomic Factors on Increased Early Childhood Blood Lead Levels: A Case Study in Chicago. *International Journal of Environmental Research and Public Health*, 21, 383.
14. **Lee, J.**, & Dessler, A. E*. (2024). Improved Surface Urban Heat Impact Assessment Using GOES Satellite Data: A Comparative Study With ERA-5. *Geophysical Research Letters*, 51(1), e2023GL107364.

Prior to 2024

15. **Lee, J*.**, & Dessler, A. E. (2023). Future Temperature-Related Deaths in the US: The Impact of Climate Change, Demographics, and Adaptation. *GeoHealth*, *7*(8), e2023GH000799.

16. **Lee, J*.**, & Dessler, A. E. (2022). The Impact of Neglecting Climate Change and Variability on ERCOT's Forecasts of Electricity Demand in Texas. *Weather, Climate, and Society*, *14*(2), 499-505.
17. **Lee, J.**, Mast, J. C., & Dessler, A. E*. (2021). The Effect of Forced Change and Unforced Variability in Heat Waves, Temperature Extremes, and Associated Population Risk in a CO_2-Warmed World. *Atmospheric Chemistry and Physics*, *21*(15), 11889-11904.
18. **Lee, J.**, Shi, Y. R., Cai, C., Ciren, P., Wang, J., Gangopadhyay, A., & Zhang, Z*. (2021). Machine Learning Based Algorithms for Global Dust Aerosol Detection from Satellite Images: Inter-Comparisons and Evaluation. *Remote Sensing*, *13*(3), 456.
19. **Lee, J.**, & Kim, K. Y*. (2018). Analysis of Source Regions and Meteorological Factors for the Variability of Spring PM10 Concentrations in Seoul, Korea. *Atmospheric Environment*, *175*, 199-209.
20. **Lee, J*.** (2017). Future Trend in Seasonal Lengths and Extreme Temperature Distributions over South Korea. *Asia-Pacific Journal of Atmospheric Sciences*, *53*, 31-41.

INVITED TALKS & PRESENTATIONS (RECENT 2 YEARS)

1. A Multiscale Model-Driven Experimental (ModEx) Framework for Urban Areas – a CROCUS approach, *American Geophysical Union Annual Meeting*, 2025
2. The 2025 CROCUS Urban Flooding and Rainfall Campaign: *International Conference on Urban Climate*, 2025
3. Urban Land Surface Temperature Downscaling in Chicago: Addressing Ethnic Inequality and Gentrification, *American Geophysical Union Annual Meeting*, 2024
4. Urban Land Surface Temperatures: Importance, Measurements, and Multidisciplinary Applications, *Florida State University*, 2024 [*Invited*]
5. Urban Land Surface Temperatures: Importance, Measurements, and Multidisciplinary Applications, *University of Illinois Chicago*, 2024 [*Invited*]
6. Urban Land Surface Temperature Downscaling in Chicago: Addressing Socioeconomic Disparities, *Seoul National University*, 2024 [*Invited*].

AWARDS & SCHOLARSHIPS

- Top Cited Paper in GeoHealth, AGU, 2025
- Outstanding Graduate Student Research Award, Texas A&M University, 2021
- Outstanding Graduate Student Seminar Award, Texas A&M University, 2021
- Best Thesis Award, Seoul National University, 2017
- Merit-Based Scholarship, Seoul National University, 2011-2017

PYTHON PACKAGES, BOOKS & EDUCATION MATERIALS

- STELAR (Spatio-TEmporaL lAg-based Regression) GAM: https://github.com/jangholee92/stelar_gam
- Python and Statistics for Climate and Atmospheric Informatics (Book - in Prep): https://github.com/jangholee92/pythonForCI
- E3SM & ELM Documentation with OLMT: https://github.com/jangholee92/ELM_Tutorial

TECHNICAL PROFICIENCY

- Proficient in Python, R, and Linux
- Proficient in ML, DL, and HPC Modules: TensorFlow, Keras, Scikit-Learn, XGBoost, Dask
- Proficient in ML and DL models: XGBoost, RF, ANN, LSTM, GAN, Diffusion Models, Autoencoders
- Experience in land-atmospheric modelling (ELM)

RESIDENTIAL STATUS

- Lawful Permanent Resident (LPR) of the United States
- South Korea Citizen